



EASTERN MEDITERRANEAN UNIVERSITY
FACULTY OF ENGINEERING
DEPARTMENT OF MECHANICAL ENGINEERING

Course Code: MENG442
Course Name: Solar Energy Engineering
Semester/ Year: Spring 2025-2026
Instructor: Assoc. Prof. Dr. Murat Özdenefe
Assistant: Mr. Hussain Shawish
Assessment: Lab
Due Date: 12th of June 2026 at 23:59 hrs.

LAB NAME: Determining instant electrical and thermal power output of a PVT system

STUDENT(S) NO.: 23704019

NAME(S), SURNAME(S): TARIK TARIQ

SECTIONS:

No.	STUDENT OUTCOMES	COURSE LEARNING OUTCOMES	WEIGHT OF SECTION /100	MARKS OBTAINED	REMARKS
S-1	6	1, 3, 5, 8	10		
S-2	6	1, 3, 5, 8	10		
S-3	6	1, 3, 5, 8	10		
S-4	6	1, 3, 5, 8	15		
S-5	6	1, 3, 5, 8	35		
S-6	6	1, 3, 5, 8	20		
					Marks for report format & referencing are inclusive in the marks of each section.
TOTAL: (Out of 100)					

Course Learning Outcomes		Student Outcomes						
		1	2	3	4	5	6	7
1	Understand the characteristics of solar energy, sun-earth geometric relations and sun path/shade diagrams.	X						
2	Comprehend beam & diffuse rad., and estimation of terrestrial rad. intercepted & absorbed by surfaces.	X						
3	Have a fundamental knowledge regarding the heat transfer aspects of the solar thermal design.	X						
4	Be able to analyze flat plate & concentrating solar thermal collectors.							
5	Develop an understanding of thermal energy storage with emphasis placed on water.	X						
6	Analyze solar thermal energy component models such as collector heat exchangers and array of collectors.	X						
7	Understand and use f-chart design methods for designing typical liquid based solar heating systems.	X	X					
8	Get familiar with basics of PVs & calculate electrical capacity of a PVT system from various measurements.	X					X	
9	Be able to design and analyze various solar systems by employing contemporary simulation tools.		X					X
	Weight of Student Outcomes	H	H				L	H

Instructions:

- Report presentation:** Draw neat, labeled diagrams where necessary. Write relevant equations and write your assumptions where necessary. Be clear and specific and include units. Give explanatory notes where necessary. Carry out each step explicitly. Provide references to the data/model/equations used. References should be from the textbook or from an authentic source.
- Submission:** Upload your report as a pdf file to the "Lab Work" section in LMS within the designated time. The file uploaded should be named as [MENG442-Lab-Student ID-Student Name]. No late submissions will be accommodated.

ABSTRACT

This experiment investigated the effect of tilt angle on the electrical and thermal performance of a photovoltaic-thermal (PVT) system. The PVT panel was tested at tilt angles of 15°, 30°, 45°, and 60° by measuring voltage, current, inlet and outlet water temperatures, and solar radiation. The highest electrical power was obtained at 15° with a value of 342.55 W, while the highest thermal power was also obtained at 15° with a value of 0.3823 kW. The electrical efficiency increased with tilt angle and reached a maximum value of 33.52% at 60°, while the highest thermal efficiency was 29.3% at 15°. The theoretical optimum tilt angle for Famagusta on 4 June at 1:30 PM was calculated as approximately 12.29°, which is close to the experimental angle that produced the highest power output.

CONTENTS

LAB NAME: Determining instant electrical and thermal power output of a PVT system	1
STUDENT(S) NO.: 23704019 NAME(S), SURNAME(S): TARIK TARIQ.....	1
ABSTRACT	2
INTRODUCTION.....	7
THEORY	8
APPARATUS.....	8
PROCEDURE	10
MEASUREMENTS, CALCULATIONS AND RESULTS.....	12
DISCUSSION AND CONCLUSION.....	16
References	18
Use of artificial intelligence	18

LIST OF GRAPHS

Graph 1 Efficiency against tilt angle 16

Graph 2 Power against tilt 17

LIST OF TABLES

Table 1 Data collected	12
Table 2 electric power values	13
Table 3 thermal power values	13
Table 4 electric efficiency	14
Table 5 thermal efficiency	14
Table 6 data for best tilt angle.....	15

LIST OF FIGURES

Figure 1 YE-1035 Thermal and Solar Cell (PVT) Training Set 9

Figure 2 Lux measuring device..... 9

INTRODUCTION

The increasing global demand for renewable and sustainable energy sources has led to significant interest in solar energy technologies. Among these technologies, photovoltaic-thermal (PVT) systems have gained attention because they combine electrical power generation and thermal energy collection within a single integrated system. Unlike conventional photovoltaic panels that convert only a portion of solar radiation into electricity while the remaining energy is lost as heat, PVT systems utilize this excess heat for useful thermal applications such as water heating.

In this experiment, the YE-1035 Thermal and Solar Cell (PVT) Training Set was used to investigate the performance of a photovoltaic-thermal system under different tilt angles. The experiment focused on determining the instantaneous electrical and thermal power outputs of the PVT system by measuring voltage, current, inlet and outlet water temperatures, water tank temperature, and solar radiation intensity.

The tilt angle of a solar collector plays an important role in determining the amount of solar radiation received by the panel surface. By varying the tilt angle, the efficiency and performance of both the photovoltaic and thermal components of the system can be analyzed. The experiment therefore provided an understanding of the relationship between solar radiation, thermal energy transfer, and electrical energy generation in a hybrid solar energy system.

The main objectives of the experiment were:

- To determine the instantaneous electrical power output of the PVT system
- To determine the instantaneous thermal power output of the system
- To analyze the effect of different tilt angles on the performance of the PVT collector
- To study the simultaneous production of electrical and thermal energy from solar radiation

THEORY

In Experiment 02, the effect of tilt angle on the electrical power output of the PVT panel is investigated by measuring the voltage, current, and solar radiation at different panel angles. The tilt angle changes the amount of solar radiation incident on the collector surface; therefore, it influences both the generated electrical power and the electrical efficiency. The electrical power is obtained from the product of the measured voltage and current, while the efficiency is found by comparing the useful electrical output with the solar energy incident on the tilted collector area. The total radiation on the tilted surface includes beam radiation, diffuse sky radiation, and ground-reflected radiation, as given in the Experiment 02 equations of the MENG442 Experiment Manual, page 2.

Equations

$$P = V I_c (W),$$

$$G_T = G_b R_b + G_d ((1 + \cos \beta) / 2) + G \rho_g ((1 - \cos \beta) / 2),$$

$$\eta = P_{\text{electric/thermal}} / (G_T A),$$

$$\eta (\%) = [P_{\text{electric}} / (G_T A)] \times 100$$

Where: P is electrical/thermal power, V is voltage, I_c is current, G_T is total radiation on the tilted surface, G_b is beam radiation, G_d is diffuse radiation, R_b is the beam radiation tilt factor, ρ_g is ground reflectance, β is panel tilt angle, A is collector area, and η is electrical efficiency.

APPARATUS

The experiment was conducted using the YE-1035 Thermal and Solar Cell (PVT) Training Set manufactured by Deneysan. The apparatus integrates both photovoltaic and thermal energy systems for the analysis of electrical and thermal performance under solar radiation.

The setup consisted of a photovoltaic (PV) panel connected to a thermal water heating system, allowing simultaneous generation of electrical and thermal energy. The training set included a control and measuring panel equipped with DC and AC voltmeters, ammeters, a regulator, switches, electrical loads, and temperature measurement sensors.

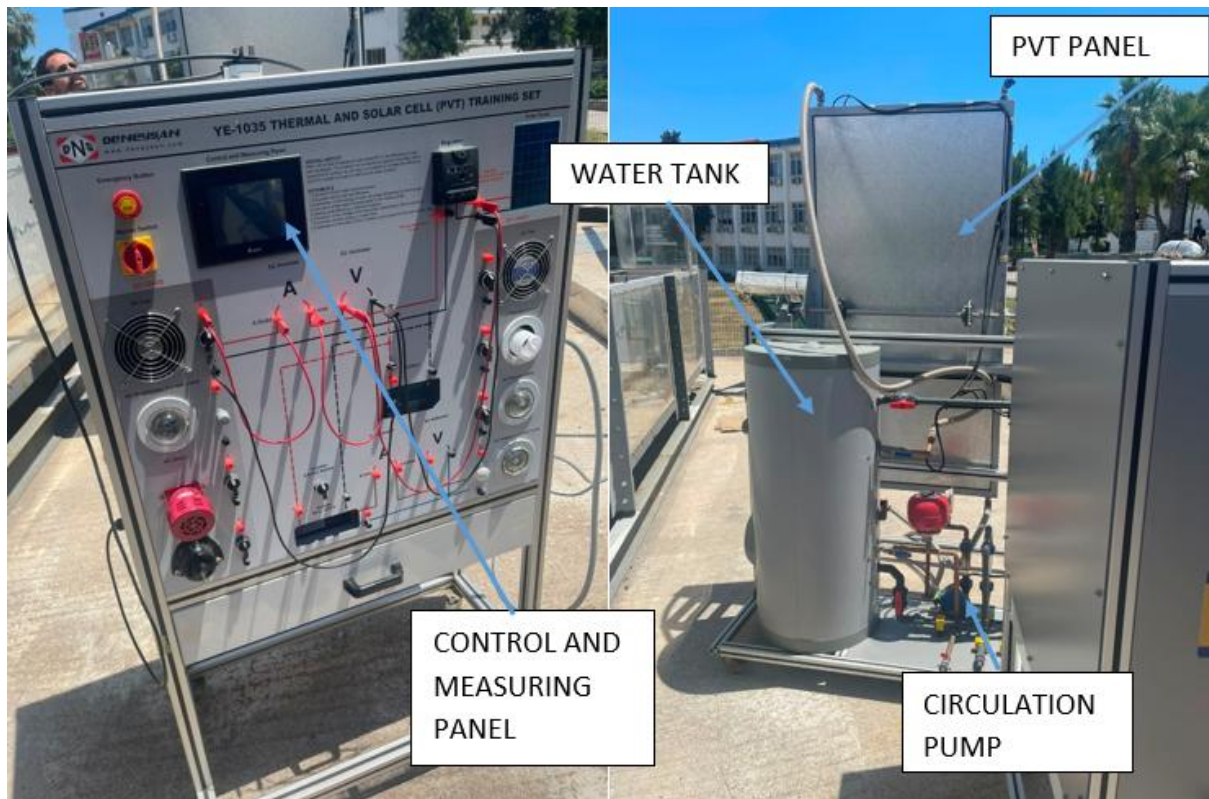


Figure 1 YE-1035 Thermal and Solar Cell (PVT) Training Set

We also need to measure the radiation falling on the tilted surface. A lux measuring device was used and a conversion factor had to be applied in order to obtain the radiation values at different tilt angle.



Figure 2 Lux measuring device

PROCEDURE

Experimental Procedure:

1. The YE-1035 Solar Cell and Hot Water (PVT) Training Set was assembled and connected according to the manufacturer's instructions.
2. The water tank was filled with sufficient water, and all pipe connections were checked to ensure there was no leakage in the system.
3. The PVT panel was positioned facing south and a fan was used as a load center to measure current and voltage.
4. The photovoltaic-thermal (PVT) panel was positioned at the first selected tilt angle.
5. The circulation pump was activated to allow water to flow through the thermal collector section of the PVT system.
6. The instruments and sensors connected to the setup were checked and calibrated where necessary.
7. The inlet water temperature, (T_{in}), was measured and recorded using the temperature sensor at the collector inlet.
8. The outlet water temperature, (T_{out}), was measured and recorded using the temperature sensor at the collector outlet.
9. The water tank temperature, ($T_{water-tank}$), was measured and recorded.
10. The electrical output current of the solar panel was measured using the appropriate measuring device and recorded.
11. The electrical output voltage of the solar panel was measured and recorded.
12. The solar radiation intensity incident on the tilted panel surface was measured using the UT83 lux meter and recorded in lux then later converted to incident radiation using a conversion factor of $1/122 \text{ W/m}^2$ per lux.
13. The instantaneous electrical power output of the PVT system was determined using:

$$P=IV$$

where:

- $(P_{\text{electrical}})$ = electrical power output (W)
- (V) = voltage (V)
- (I) = current (A)

The values for voltage and current were obtained and recorded after each iteration.

14. The instantaneous thermal power output of the PVT system was determined using:

$$P_{\text{Thermal}} = \dot{m} c_p (T_{\text{out}} - T_{\text{in}})$$

where:

- (P_{Thermal}) = thermal power output (W)
- $(\dot{m})$ = mass flow rate of water (kg/s)
- (c_p) = specific heat capacity of water (J/kg·K)
- (T_{out}) = outlet water temperature (°C)
- (T_{in}) = inlet water temperature (°C)

Outlet temperature and inlet temperatures were recorded for each iteration at different tilt angle.

15. The experiment was repeated for three iterations at different tilt angles of the PVT panel.

16. For each tilt angle, all measured values including current, voltage, inlet temperature, outlet temperature, water tank temperature, and lux readings were recorded carefully in tabular form.

17. After completing all measurements, the light source and circulation pump were switched off, and the apparatus was allowed to cool down before shutdown.

MEASUREMENTS, CALCULATIONS AND RESULTS

The following table represents values collected during the experiment which was done at different tilt angle. These values were used to calculate electrical and thermal power.

Table 1 Data collected

	15°	30°	45°	60°
Voltage (v)	40.3	40	39.8	39.3
Current (A)	8.5	8.2	8	7.12
T _{in}	36.6	38	39.3	40.1
T _{out}	37.7	38.6	39.8	40.6
lux	120000	113000	98700	76800
T _{water tank}	36	36.1	36.3	36.6

$$A = 1.326 \text{ m}^2$$

Conversion from lux to Incident radiation:

$$1 \text{ W/m}^2 = 122 \text{ lux}$$

$$\text{Sample calculation @15° tilt angle: } 120,000 * 1/122 \text{ w/m}^2 = \mathbf{983.61 \text{ w/m}^2},$$

$$113000 \text{ lux @30°} = \mathbf{926.23 \text{ w/m}^2}$$

$$98700 \text{ lux @45°} = \mathbf{809.01 \text{ w/m}^2}$$

$$76800 \text{ lux @60°} = \mathbf{629.508 \text{ w/m}^2}$$

Electrical power calculations:

Table 2 electric power values

Current (A)	Voltage (V)	Power (W)
8.5	40.3	342.55
8.2	40	328
8	39.8	318.4
7.12	39.3	279.816

Thermal power calculations:

Mass flow rate computation: $\dot{m} = \rho \dot{V} = 1000 \text{ kg/m}^3 * 300 \text{ L/h} * 0.001 \text{ m}^3/\text{L} * 1/3600 \text{ h/s} = 0.083 \text{ kg/s}$

Table 3 thermal power values

$T_{in} (^{\circ}\text{C})$	$T_{out} (^{\circ}\text{C})$	$\dot{m} (\text{kg/s})$	$c_p (\text{KJ/kg.k})$	Power (KW)
36.6	37.7	0.083	4.187	0.3823
38	38.6	0.083	4.187	0.2085
39.3	39.8	0.083	4.187	0.1738
40.1	40.6	0.083	4.187	0.1738

Sample calculation @15° tilt angle: $\text{power}_{\text{thermal}} = \dot{m} c_p (T_{out} - T_{in}) = 0.083 * 4.187 * (37.7 - 36.6) = 0.3823 \text{ KW}$

Electrical efficiency calculations:

$$\eta = P/IA$$

sample calculation of electrical and thermal efficiency @15° tilt angle:

$$\eta_{\text{electrical}} = 342.55 \text{ W} / 983.61 \text{ W/m}^2 * 1.326 \text{ m}^2 = 0.2626 = \mathbf{26.3\%}$$

Table 4 electric efficiency

P(w)	I (w/m ²)	A(m ²)	η
342.55	983.61	1.326	0.2626
328	926.23	1.326	0.2671
318.4	809.01	1.326	0.2968
279.816	629.508	1.326	0.3352

thermal efficiency calculations:

$$\eta = P_{\text{thermal}}/IA$$

sample calculation of electrical and thermal efficiency @15° tilt angle:

$$\eta_{\text{electrical}} = 382.3\text{w}/983.61 \text{ w/m}^2 * 1.326 \text{ m}^2 = 0.293 = \mathbf{29.3\%}$$

Table 5 thermal efficiency

P(w)	I(w/m ²)	A(m ²)	η
382.3	983.61	1.326	0.293
208.5	926.23	1.326	0.1697
173.8	809.01	1.326	0.162
173.8	629.508	1.326	0.208

Best tilt angle using analytical method:

Famagusta, Cyprus - 4 June at 1:30 PM Local Time

This calculation uses the solar-angle notation and equations from the MENG442 Solar Radiation lecture notes. The PVT panel is assumed to face due south, so the surface azimuth angle is zero. All angles are in degrees.

Input Data and Intermediate Results

Table 6 data for best tilt angle

Quantity	Symbol	Value	Comment
Location	-	Famagusta, Cyprus	Given location
Latitude	ϕ	35.12° N	Positive for northern hemisphere
Longitude	Lloc	33.9° E	Local meridian of Famagusta
Date	-	4 June	Calculation date
Day number	n	155	June: $n = 151 + i$, with $i = 4$
Local clock time	-	13:30	Cyprus daylight saving time
Standard time used	-	12:30	Local clock time minus 1 h for daylight saving
Standard meridian	Lst	30° E	Famagusta standard meridian from the lecture notes
Equation of time	E	≈ 1.9 min	Value for 4 June
Solar time	-	≈ 12.792 h	Corrected time used in solar-angle relations
Declination angle	δ	$\approx 22.54^\circ$	From the Cooper declination equation in the notes
Hour angle	ω	$\approx 11.88^\circ$	Afternoon is positive
Surface azimuth angle	γ	0°	Due south-facing surface

Best Tilt Angle

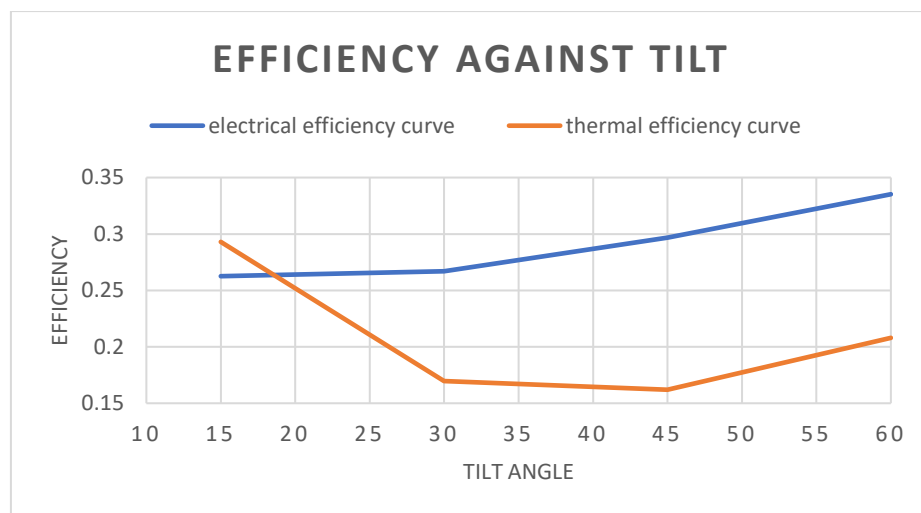
$$\beta_{\text{opt}} = \phi - \tan^{-1} (\tan \delta / \cos \omega) = 35.12^\circ - \tan^{-1} [\tan (22.42^\circ) / \cos (11.88^\circ)] = 12.29^\circ \approx 12^\circ\text{--}13^\circ$$

Final recommended tilt angle: 12° to 13° from horizontal

This is the instantaneous optimum tilt for the stated date and time when the PVT panel is fixed facing due south. The result is lower than the tracking tilt because the sun is west of south at 1:30 PM solar-time-corrected conditions.

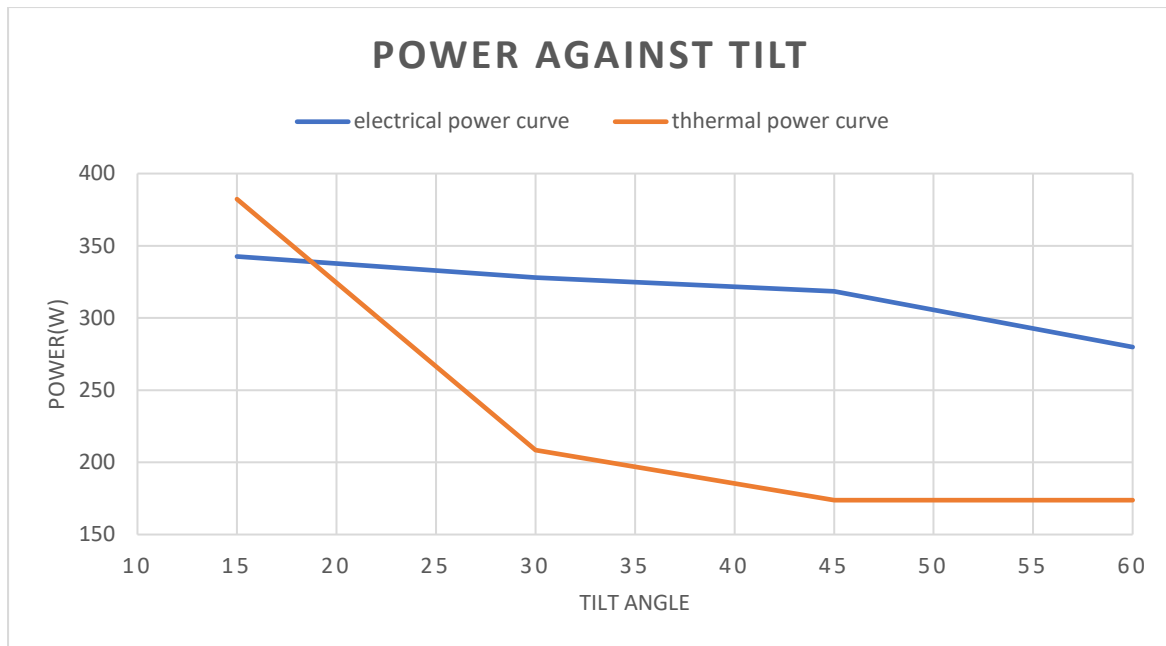
DISCUSSION AND CONCLUSION

The below figure shows the graph of efficiency against tilt. We can see from the graph when the tilt angle was changed from 15 to 30 degrees there was a steep fall in the efficiency of the thermal side of the system while electrical efficiency increased. At around 18.5 degrees we can spot both efficiency curves intersecting each other.



Graph 1 Efficiency against tilt angle

We have a similar case with the electrical power curve. We can see the thermal power curve steeply going down while electrical power drops as well from 15° to 30° but not at the same rate as the thermal power curve. Both curves intersect each other at 18.5° tilt angle.



Graph 2 Power against tilt

At the intersection point of approximately **18.5°**, the thermal and electrical efficiencies (and power outputs) of the PVT system are equal. Below this tilt angle, the system is thermally dominant meaning it delivers more heat than electricity while above it, electrical output overtakes thermal. Notably, this crossover occurs very close to the theoretically optimal tilt of **12.29°**, which means that even at peak performance the system is still operating in the thermally dominant region. This behavior is physically explained by the high solar altitude on June 5th in Famagusta ($\delta = 22.54^\circ$): at shallow tilts near optimal, the high irradiance striking the collector surface drives strong thermal absorption, but simultaneously heats the PV cells, suppressing their electrical efficiency. As the tilt increases beyond 18.5°, incident irradiance decreases, the cells cool down, and electrical efficiency gradually recovers and surpasses thermal efficiency making the intersection point a clear fingerprint of the inherent thermal-electrical trade-off characteristic of PVT systems operating under intense summer irradiance.

References

- [1] National Oceanic and Atmospheric Administration, "NOAA Solar Calculator," NOAA Global Monitoring Laboratory. Accessed: Jun. 11, 2026. [Online]. Available: <https://gml.noaa.gov/grad/solcalc/>
- [2] J. A. Duffie and W. A. Beckman, *Solar Engineering of Thermal Processes*, 4th ed. Hoboken, NJ, USA: John Wiley & Sons, 2013. doi: 10.1002/9781118671603.

Use of artificial intelligence

Students are required to explicitly state the use of AI-generated text, data analysis, visuals, or other content in their report under the heading "Use of Artificial Intelligence", which should be placed immediately after the "References" section. This statement must clearly specify the AI tools used (e.g., ChatGPT, Midjourney, Gemini) and the purpose for which they were employed (e.g., generating preliminary text for a literature review, data analysis).

Example Statement: "In this study, [XYZ AI tool] was used for [purpose, e.g., generating preliminary text for a literature review]."

AI-generated content must not exceed 20% of the total word count of the report.

For details, please visit: <https://grad.emu.edu.tr/en/academic-issues/use-of-ai-in-thesis>

S. No.	AI Tool Used	Section	Section No.	Page No.	Purpose of use
1	ChatGPT	Introduction	1	1	Used to help find references and refining the introduction section
2	DeepSeek	Theory	2	3-5	Used to generate information about the theory/models
3	ChatGPT	Apparatus	3	6	Used to explain more about the available apparatus for similar tests
4	ChatGPT	Procedure	4	7	Used for gathering additional information about the tests and applied standards
5	Claude	Measurements, Calculations and Results	5	8-9	Used to generate charts from the measured data as well as refine wordings
6	Claude	Discussion and Conclusion	6	10	Used for language proofing

